The residual bound, step by step

figures `fig_kernel_free.png`, `fig_design_range.png`; scripts listed in the Reproduction section

Symbols

$\tau,\ \tau_{\text{end}}$	time since onset; window end	x_{end}	$= C_2\tau_{\text{end}}$, dimensionless window
y_i	measured rate at τ_i	N	number of samples in the window
C_1, C_2	nominal amplitude $K\dot{M}$ and exponent	$K,\ \dot{M}$	calibrated gain; ramp rate
$W,\ z_{CoM},\ J_P$	weight, CoM height, pivot inertia	Wz_{CoM}	$= J_P C_2^2$ (the identity)
$e_\omega(\tau)$	true rate minus the nominal	$b,\ n$	gyro bias; disturbance
C	fitted baseline (pre-onset median)	r_i	residual $y_i - \hat{\omega}_i$
$\delta t_c,\ \beta$	onset shift of the witness branch; its sinh coefficient, $\tanh(C_2\delta t_c) = -\beta/C_1$	δe_ω	what the shift cannot absorb
$\rho(s)$	deviation forcing (small-angle + GE)	$\bar{\rho}$	window average bound on ρ, eq. (91)
$\dot{\rho}$	its time derivative	$\varphi_{\max}$	5° excitation tilt cap
$\omega_{\max}$	end-rate anchor, eq. (16) block	$l_{\text{arm}},\ \beta_M$	pivot arm; GE coefficient
ΔM_{win}	in-window moment excursion	$E(\tau)$	envelope $\bar{\rho}\sinh(C_2\tau)/(J_P C_2)$, eq. (90)
n_{hi}	residual content above 5 Hz	$\kappa_q,\ s_{\text{med}}$	quiet-window shape ratio; median in-window enhancement (1.60)
$\hat{n}$	noise estimate, Sec. 6		

1 The measurement and the fit

The record in the window and the fitted model are

$$y_i = C_1\big(\cosh(C_2\tau_i) - 1\big) + e_\omega(\tau_i) + b + n_i, \tag{1}$$

$$\hat{\omega}_i = C_1^*\big(\cosh(C_2^*(\tau_i - \delta t_c^*)) - 1\big) + C, \tag{2}$$

and on the pre-onset segment the baseline matches the bias,

$$C \approx b. \tag{3}$$

2 The residual, and what the shift absorbs

Subtracting (2) from (1) with (3),

$$r_i \;=\; y_i - \hat{\omega}_i \;=\; \underbrace{C_1\big(\cosh(C_2\tau_i) - 1\big) + e_\omega(\tau_i)}_{\text{to be absorbed}} \;-\; C_1^*\big(\cosh(C_2^*(\tau_i - \delta t_c^*)) - 1\big) \;+\; n_i. \tag{4}$$

The question is how much of the braced part a member of the family can match. Why a sinh term, of all shapes? Two facts meet. First, e_ω *grows* like sinh: its envelope (90) is $E(\tau) \propto \sinh(C_2\tau)$, because the deviation rides the instability's own growing mode — so peeling off a sinh multiple targets the dominant part of the error. Second, sinh is exactly what the family can absorb: it is the tangent direction of the onset shift, $\partial_{\delta t_c} C_1(\cosh(C_2(\tau - \delta t_c)) - 1)\big|_0 = -C_1C_2\sinh(C_2\tau)$, so adding $\beta\sinh$ lands inside the family (made exact in (6)–(7)). Had the envelope been, say, τ^2-shaped, subtracting $\beta\tau^2$ would target the error just as well and help not at all — the family could not represent it. The coincidence of the two — the error's dominant shape *is* the direction the fit can absorb — is the entire reason the measured residuals sit flat. Add and subtract that one sinh term — this *defines* δe_ω, and the equality holds at every τ, both window ends included:

$$C_1\big(\cosh(C_2\tau) - 1\big) + e_\omega(\tau) \;=\; \underbrace{C_1\big(\cosh(C_2\tau) - 1\big) + \beta\sinh(C_2\tau)}_{\hat{h}(\tau),\ \text{the witness}} \;+\; \delta e_\omega(\tau). \tag{5}$$

The witness is a member of the family in disguise. With the addition identity

$$\cosh\big(C_2(\tau - \delta t_c)\big) = \cosh(C_2\delta t_c)\cosh(C_2\tau) - \sinh(C_2\delta t_c)\sinh(C_2\tau), \tag{6}$$

matching the cosh and sinh coefficients [$R\cosh(C_2\delta t_c) = C_1$, $-R\sinh(C_2\delta t_c) = \beta$] gives, *exactly*,

$$\hat{h}(\tau) = R\big(\cosh(C_2(\tau - \delta t_c)) - 1\big) + (R - C_1), \qquad R = \sqrt{C_1^2 - \beta^2}, \quad \tanh(C_2\delta t_c) = -\beta/C_1, \tag{7}$$

a branch with amplitude R, onset δt_c, and a baseline offset $R - C_1$ — three numbers, all inside the family's freedoms in (2) (the offset is absorbed by C: the witness's baseline slot carries $b + (R - C_1)$, not b alone, and the "≈" of (3) is exactly this second-order constant; what it costs, and on which segment it lands, is settled in the remark closing the pre-onset section). On the campaign $\beta/C_1 \le 0.023$, so the amplitude and baseline adjustments are second order ($|R/C_1 - 1| \le 0.03\%$); the shift carries the first-order content. The remainder is

$$\delta e_\omega(\tau) \;=\; e_\omega(\tau) - \beta\sinh(C_2\tau). \tag{8}$$

3 Choosing the shift: match the true curve at the window end

One free number, δt_c (equivalently β), and one choice: make the shifted branch meet the true curve at τ_{end}. Setting $\delta e_\omega(\tau_{\text{end}}) = 0$ in (8),

$$\beta \;=\; \frac{e_\omega(\tau_{\text{end}})}{\sinh(C_2\tau_{\text{end}})}, \qquad \delta t_c \;=\; -\frac{1}{C_2}\,\operatorname{artanh}\Big(\frac{\beta}{C_1}\Big) \;\approx\; -\frac{e_\omega(\tau_{\text{end}})}{C_1 C_2\sinh(C_2\tau_{\text{end}})}. \tag{9}$$

Since $e_\omega > 0$ (the true curve rises faster than the nominal), $\delta t_c < 0$: the branch starts *earlier* than the true onset. On the campaign $\beta/C_1 = 0.010$–0.023, i.e. $|\delta t_c| = 2$–4 ms.

Admissibility — why $|\beta| < C_1$ is guaranteed. (7) needs R real, i.e. $|\beta| < C_1$; otherwise the witness is not a cosh branch at all. This is provable, not assumed: the envelope (90) gives $|e_\omega(\tau_{\text{end}})| \le \bar{\rho}\sinh(C_2\tau_{\text{end}})/(J_P C_2)$, so $|\beta| \le \bar{\rho}/(J_P C_2)$, and dividing by $C_1 = \dot{M}/W z_{CoM}$ with the identity $J_P C_2^2 = W z_{CoM}$,

$$\frac{|\beta|}{C_1} \;\le\; \frac{\bar{\rho}\, C_2}{\dot{M}} \;\triangleq\; u. \tag{9a}$$

So $|\beta| < C_1$ holds whenever $\bar{\rho} < \dot{M}/C_2$ — *the ramp must sweep more moment in one e-folding time $1/C_2$ of the instability than the window-averaged remainder.* (An *e-folding* is the natural unit of an exponential process: the interval over which $e^{C_2\tau}$ grows by a factor e, i.e. $1/C_2 \approx 0.2$ s here — the base-e analogue of a half-life. $x_{\text{end}} = C_2\tau_{\text{end}}$ counts how many such units the window spans; this document measures time in $1/C_2$ throughout, because in that unit the same formulas give the same numbers for every vehicle and ramp rate.) A protocol that violated this would have its excitation comparable to the model error it is meant to dominate, so any meaningful design satisfies it with room: $u \le 0.16$ at the campaign's slowest ramp, ≤ 0.14 at the worst design-box corner of Sec. 9. The deviation itself sits well inside: on the exact nonlinear solution $e_\omega(\tau_{\text{end}}) \le 0.29\, E(\tau_{\text{end}})$ at every rate. (The *fitted* $\beta/C_1 = 0.010$–0.023 carries noise-driven scatter on top of the e_ω part, so it is not the quantity the envelope governs run by run — but either number is two orders below the admissibility limit 1.) The same u is the argument of the $\operatorname{artanh}(u)/u$ factor in (21): one dimensionless number controls both the witness's admissibility and the exactness of the shift ceiling.

What if $u \ge 1$ — ramps slower than $\bar{\rho}C_2$. Three things, in decreasing order of certainty. (i) Only the *a priori* guarantee is lost: (9a) is sufficient, not necessary, so the witness may well still be admissible — the realised β/C_1 sits some thirty times below the envelope ceiling on the campaign — but nothing proves it before the experiment any more. (ii) If $|\beta| \ge C_1$ did occur, the combination $C_1\cosh(C_2\tau) + \beta\sinh(C_2\tau)$ is no longer a cosh branch: it equals $\sqrt{\beta^2 - C_1^2}\,\sinh\big(C_2\tau + \varphi\big)$ (a pure exponential at $|\beta| = C_1$), and no real time shift of the family (2) draws a sinh-type curve. Step (10) then has no admissible competitor to hand to the minimiser, and the cap loses its proof — even though the Dirichlet bound (11)–(17) on $e_\omega - \beta\sinh(C_2\tau)$ itself survives, since it never used admissibility. Moving the matching point earlier does not rescue it: the envelope of e_ω and $\sinh(C_2\tau)$ grow in the same ratio, so the required β ceiling is the same at every matching point. (iii) Physically, $\dot{M} < \bar{\rho}C_2$ says the moment swept in one e-folding of the instability no longer dominates the model remainder — the identification concept itself, not just its certificate, is degrading; the campaign's slowest ramp runs at $u \le 0.16$, comfortably inside. Numerically, $u = 1$ is reached at $\dot{M} = \bar{\rho}C_2 \approx 0.016$ N m/s for the campaign vehicle and at $\dot{M} = 0.015$ N m/s at the worst design-box corner of Sec. 9 ($m = 3.22$ kg, arm 0.160 m); a hypothetical $\dot{M} = 0.01$ N m/s ramp would have $u \approx 1.5$. So (9a) doubles as an analytical *lower* rate limit for the protocol, $\dot{M} > \bar{\rho}C_2 \approx 0.015$ N m/s — a factor ∼7 below the slowest campaign rate — complementing the allocator-saturation *upper* limit on $\dot{M}$. In practice the logic runs forward, and this is the intended reading: the guarantee fails *only* there, and since $\bar{\rho}$ and C_2 need nothing but the coarse design-box ranges known before any experiment, one computes the box-worst $\bar{\rho}C_2 = 0.015$ N m/s in closed form first and then schedules the ramp rates above that floor with margin — the campaign's $\dot{M} \geq 0.10$ N m/s is a factor ≈ 6.6 above it, giving $u \leq 0.14$ everywhere in the box.

Could the family be composed so the problem never arises? Partly, and where it splits is instructive. Replace the shifted-cosh family (2) by its linear span, fitting $\hat{\omega} = A(\cosh C_2\tau - 1) + B \sinh C_2\tau + C$: the witness $\omega_{\text{nom}} + \beta \sinh(C_2\tau)$ is then a member for *every* β (take $A = C_1$, $B = \beta$, $C = b$), so the argmin step (10) holds unconditionally and admissibility disappears from the residual cap altogether. The added column is no stranger: sinh is exactly the tangent direction of the time shift ($\partial_{\delta t_c}$ of the branch at $\delta t_c = 0$ is $-C_1C_2 \sinh$, Appendix A.2), so this is the linearised version of the same estimator. The cost is one more degree of freedom in the noise term (minor) and a near-collinear pair of columns — cosh -1 and sinh agree to $O(e^{-2x_{\text{end}}})$ at the window end — so the residual stays clean while the individual coefficients A, B get noisy: the ridge of Sec. 4, resurfacing as conditioning. What *cannot* be designed away is the readout. Converting (A, B) back into an onset time needs $R = \sqrt{A^2 - B^2}$ real, i.e. $|B| < A$ — the very same condition, now appearing where it belongs. $u < 1$ is not an artifact of the cosh parametrisation; it is the statement that the data actually describes a delayed cosh onset rather than a sinh-type curve, and no reparametrisation makes data interpretable that is not. The practical split: the linear family buys an *unconditional* residual cap, and $u < 1$ then governs only whether the fitted curve has a physical onset to read off.

Then, because the minimiser in (2) is at least as good as this one particular shifted branch,

$$\boxed{\sum_{i=1}^{N} r_i^2 \;\leq\; \sum_{i=1}^{N} \big(\delta e_\omega(\tau_i) + n_i\big)^2 \qquad \Longrightarrow \qquad \text{RMS}(r) \;\leq\; \text{RMS}(\delta e_\omega) + \text{RMS}(n).} \tag{10}$$

Everything now rests on one question: **how big is δe_ω?**

4 In exactly what sense the true curve is a time-shifted nominal

The ansatz of (5) deserves a precise statement, because "the true model is the nominal, time-shifted" is true in one sense, exactly, and false in another, and the gap between the two *is* δe_ω.

The local (osculating) sense — exact at every instant. Splitting the manuscript's Duhamel kernel (89) with the addition identity gives the deviation two running coefficients,

$$e_\omega(\tau) = A(\tau)\cosh(C_2\tau) - B(\tau)\sinh(C_2\tau), \quad A = \frac{1}{J_P}\int_0^\tau \cosh(C_2 s)\rho\, ds, \quad B = \frac{1}{J_P}\int_0^\tau \sinh(C_2 s)\rho\, ds, \tag{10a}$$

so the true structural curve is $(C_1{+}A)\cosh(C_2\tau) - B\sinh(C_2\tau) - C_1$. Freezing the coefficients at an instant τ and applying the shift identity (6) backwards, the curve agrees there — in value *and* slope — with exactly one branch, the *osculating branch*, whose parameters are

$$R(\tau) = \sqrt{\big(C_1{+}A(\tau)\big)^2 - B(\tau)^2}, \qquad \tanh\big(C_2\,\delta t_{\text{loc}}(\tau)\big) = \frac{B(\tau)}{C_1 + A(\tau)}. \tag{10b}$$

At the onset $A = B = 0$, so $R = C_1$ and $\delta t_{\text{loc}} = 0$: **the true curve starts as the nominal, exactly.** As the window progresses the osculating parameters drift monotonically (measured on the exact tip-over: $\delta t_{\text{loc}} = 0$ at the onset, still only 2.5/0.4/0.1 ms at mid-window for $\dot{M} = 0.10/0.45/1.20$, then 546/239/85 ms at the window end, with $R(\tau_{\text{end}})/C_1 = 13.0/3.1/1.5$). The late blow-up is structural: e_ω's tail grows like $e^{2C_2\tau}$, faster than any branch, so the branch that hugs it must run away in amplitude and shift together — and those two parameters are ridge-degenerate, wildly different $(R, \delta t)$ pairs drawing nearly the same curve over the window. This is the same degeneracy that makes $(C_1^*, C_2^*, \delta t_c^*)$ unidentifiable from the residual.

The window (L^2) sense — the estimator's sense. No single branch can follow the drifting profile (10b); a fit must pick one point of it. The least-squares stationarity $\langle r, \sinh\rangle = 0$ weights the choice by $\sinh^2(C_2\tau)$, which concentrates 63% of its mass in the last e-folding $1/(2C_2) \approx 90$–140 ms of the window (86% in the last two), so the fitted single shift is essentially the end-matching one of (9) — and it is *small*: $\beta/C_1 = 1.0$–2.3%, i.e. $|\delta t_c| = 2$–4 ms on the campaign. In this sense the true curve *is* a slightly-shifted nominal: one branch, shifted by milliseconds, matches it over the whole window up to exactly δe_ω — which vanishes at both ends and sags mid-window where the drifting profile is farthest from its end value. That sag is what (11)–(17) bound.

One sign remark connects the two senses. The osculating description carries a *larger* amplitude with a *later* onset ($\delta t_{\text{loc}} > 0$); the witness of (9) carries the nominal amplitude with an *earlier* onset ($\delta t_c < 0$). Along the ridge these describe nearly the same curve — larger-and-later $\approx$ same-and-earlier — and the deployed estimator, whose amplitude is pinned to $K\dot{M}$, realises the earlier-onset representative: the sign observed on the campaign, and the sign whose $M_{\rm crit}$ cost (21) charges.

Remark — a perfect fit at all N samples is impossible, and that is not a defect of the fit. Two reasons, one counting and one dynamical. Counting: a branch has four free numbers $(C_1^*, C_2^*, \delta t_c^*, C)$, so it can be made to pass through at most four chosen samples; the other $N-4$ dimensions of the sample space are out of its reach, and e_ω has components there. Dynamical, and stronger: if a branch agreed with the true curve on any subinterval, then $\delta e_\omega \equiv 0$ there, and (13) would force $\dot\rho \equiv 0$ on that subinterval — the true curve solves the *forced* equation, the branches solve the homogeneous one, and they can coincide on an interval only where the forcing vanishes. Exact agreement at $\tau_1, \ldots, \tau_N$ is therefore not "difficult"; for $\dot\rho \not\equiv 0$ it is unattainable in principle.

This is why the section's claim is framed as it is. "The true curve is a time-shifted nominal" cannot be asserted exactly; asserted with the remainder, it *cannot be wrong*: eq. (5) is an identity — δe_ω is defined as whatever the shifted branch misses — and the entire scientific content lives in one place, the bound (17) on that remainder. The structure is that of Taylor's theorem: "f is approximately a polynomial" is a hope, "f equals the polynomial plus a remainder obeying a stated bound" is a theorem. Here the remainder is 0.16–0.17°/s realised, capped by (17) at 0.5–0.9°/s from pre-experiment constants, and located — zero at both ends, sagging mid-window — so the imperfection of the time-shift description is quantified in size, ceiling and shape.

The same statement has a clean linear-algebra reading: the model family is a four-parameter manifold in the N-dimensional sample space, the true record carries a component transverse to it that no choice of parameters can reach, and δe_ω is exactly that transverse coordinate — introduced so that the description becomes an identity rather than an approximation. The Appendix makes this picture precise and connects it to the manuscript's projector formalism.

5 How big is δe_ω: two boundary zeros and one comparison

Step 1 — δe_ω vanishes at both ends. At $\tau = 0$: $e_\omega(0) = 0$ (the deviation starts at rest) and $\sinh(0) = 0$, so $\delta e_\omega(0) = 0$. At $\tau = \tau_{\rm end}$: zero by the choice (9).

$$\delta e_\omega(0) \;=\; \delta e_\omega(\tau_{\rm end}) \;=\; 0. \tag{11}$$

Step 2 — the equation δe_ω obeys. The deviation dynamics of the manuscript are $J_P \ddot e_\varphi = W z_{CoM}\, e_\varphi + \rho$ with $e_\omega = \dot e_\varphi$. Differentiating once and using $W z_{CoM} = J_P C_2^2$,

$$\ddot e_\omega \;=\; C_2^2\, e_\omega + \dot\rho / J_P. \tag{12}$$

$\sinh(C_2\tau)$ satisfies $(\sinh)'' = C_2^2 \sinh$ — it is a solution of the same equation with zero forcing — so subtracting $\beta \sinh$ changes nothing on the right side:

$$\delta\ddot e_\omega \;=\; C_2^2\, \delta e_\omega + \dot\rho/J_P, \qquad \delta e_\omega(0) = \delta e_\omega(\tau_{\rm end}) = 0. \tag{13}$$

Step 3 — comparison (this is the whole proof). Let $\bar f(\tau) \ge |\dot\rho(\tau)|$ and let w solve the same problem with forcing $-\bar f$:

$$\ddot w = C_2^2 w - \bar f/J_P, \qquad w(0) = w(\tau_{\rm end}) = 0. \tag{14}$$

Claim: $|\delta e_\omega(\tau)| \le w(\tau)$. Proof in two lines: the difference $u = w - \delta e_\omega$ satisfies $\ddot u - C_2^2 u = -(\bar f + \dot\rho)/J_P \le 0$ with $u = 0$ at both ends — note the ≤ 0 belongs to this *combination*, at every τ, because $\bar f \ge |\dot\rho|$ makes $\bar f + \dot\rho \ge 0$. If u had a negative interior minimum, there $\ddot u \ge 0$ (the valley bottom bends upward: $\dot u = 0$ there, and $u(\tau^* {+} h) = u(\tau^*) + \frac{1}{2}\ddot u(\tau^*)h^2 + o(h^2) \ge u(\tau^*)$ for both signs of h forces $\ddot u \ge 0$) and $-C_2^2 u > 0$ (strictly, because $u < 0$ there — a minimum value ≥ 0 would give $-C_2^2 u \le 0$ and no contradiction), so $\ddot u - C_2^2 u > 0$ — contradicting the ≤ 0 that holds everywhere. Hence no negative interior minimum exists; with $u = 0$ at both ends and u continuous, $u \ge 0$, i.e. $\delta e_\omega \le w$; the same argument with $u = w + \delta e_\omega$ gives $\delta e_\omega \ge -w$. $\square$ (The picture is the parabola: between two zero endpoints, a negative dip must open *upward* at its bottom — for $u = a\tau(\tau - \tau_{\rm end})$ that forces $a > 0$, $\ddot u > 0$ — while the inequality, rewritten $\ddot u \le C_2^2 u$, makes u concave *downward* wherever $u < 0$. The two demands cannot meet at the bottom of a dip, so no dip. The sign of C_2^2 is what makes this work: for an oscillator, $\ddot u + C_2^2 u \le 0$, negative dips between zeros are allowed — sin draws one — and there is no maximum principle; the repelling, cosh-type sign of the tip-over instability is exactly what forbids them.)

Step 4 — solve (14) in closed form. For constant $\bar{f} = \sup|\dot{\rho}|$, substitution verifies

$$w(\tau) \;=\; \frac{\sup|\dot{\rho}|}{J_P C_2^2}\left(1 - \frac{\cosh\big(C_2(\tau - \frac{\tau_{\text{end}}}{2})\big)}{\cosh(x_{\text{end}}/2)}\right) \;\leq\; \frac{\sup|\dot{\rho}|}{W z_{CoM}}. \tag{15}$$

(Here $x_{\text{end}} = C_2\tau_{\text{end}}$: at $\tau = 0$ and $\tau = \tau_{\text{end}}$ the numerator equals $\cosh(x_{\text{end}}/2)$, so the bracket vanishes at both ends; it peaks at mid-window, where the bound is $(1 - \operatorname{sech}(x_{\text{end}}/2))\sup|\dot{\rho}|/W z_{CoM}$.) **No exponential appears.** The envelope (90) grows like $\sinh(C_2\tau)$ because nothing stops the unstable mode of (12); here the two boundary zeros of (11) — one free, one bought with δt_c — pin that mode down. This is what "the fit absorbs the growing part" means as an equation.

Step 5 — the forcing rate, from the same channels as $\bar{\rho}$. Differentiating the gravity remainder (84) in time gives $\dot{\rho}_{\text{grav}} = \big[W l_{\text{arm}} \sin\varphi + W z_{CoM}(\cos\varphi - 1)\big]\,\omega$, and the two bracket terms have *opposite signs*: the height term is negative and cancels against the arm term instead of adding to it. The bracket stays inside $[0,\ W l_{\text{arm}}\sin\varphi]$ as long as $l_{\text{arm}} \geq z_{CoM}\tan(\varphi/2)$ — held with a factor 3.4 over the whole design box (0.090 m against at most $0.30\tan 5^\circ = 0.026$ m) — so $W z_{CoM}$ drops from the bound *by sign, not by approximation.* This is the same cancellation $\bar{\rho}$ (91) already uses ($W z_{CoM}(\sin\varphi - \varphi) \leq 0$ there), and it is exactly what the manuscript's (85) says at $\varphi^* = 0$: $G''(0) = W(l_p + p_{\text{off}})$, the $W z_{CoM}\sin\varphi^*$ term vanishing. Bounding on the 5° box, with the *true* end-rate anchor, a priori:

$$\omega_{\max} \;=\; K\dot{M}(\cosh x_{\text{end}} - 1) \;+\; \frac{\bar{\rho}\sinh x_{\text{end}}}{J_P C_2} \;\geq\; \omega_{\text{nom}}(\tau_{\text{end}}) + e_\omega(\tau_{\text{end}})$$

by the envelope (90) — the nominal end rate alone would undercount the true rate by $\varepsilon(\tau_{\text{end}})$, so the envelope of e_ω is added outright; the correction is $1 + u\coth(x_{\text{end}}/2)$, i.e. +13% at the slowest ramp and +1.5% at the fastest with the 5° box, and every term is a pre-experiment constant. (History: a first draft used $K\dot{M}\sinh x_{\text{end}}$, an overbound by $\coth(x_{\text{end}}/2)$; a second used the nominal end rate alone, leaving its excess to the 1.05 — which now covers only the interior shape transfer.) Then:

$$\sup|\dot{\rho}| \;\leq\; \underbrace{W l_{\text{arm}}\varphi_{\max}\,\omega_{\max}}_{\dot{\rho}_2,\ \text{decays like } e^{2C_2(s-\tau_{\text{end}})}} \;+\; \underbrace{|\beta_M|\big(\dot{M}\varphi_{\max} + \Delta M_{\text{win}}\,\omega_{\max}\big)}_{\dot{\rho}_1,\ \text{decays like } e^{C_2(s-\tau_{\text{end}})}}. \tag{16}$$

Every constant is available before the experiment: W, z_{CoM} from the scale, J_P from CAD, l_{arm} from geometry, $\varphi_{\max}$ the design box, β_M from the GE model, $\dot{M}$ and the window from the protocol.

Where $\varphi_{\max}$ enters, and why 5°. $\varphi_{\max}$ is not a measurement — it is the tilt box the protocol guarantees, and it enters the theory at exactly three points: the channel amplitudes above (every trigonometric remainder is monotone in φ on $[0, \varphi_{\max}]$, so each sup is its value at the box edge — the $\varphi^* = 0$ Taylor coefficients of (85) evaluated at $\varphi_{\max}$ powers); the tilt-limited window of the design sweep ($\sinh x_{\text{end}} - x_{\text{end}} = \varphi_{\max} W z_{CoM} C_2/\dot{M}$; the campaign check uses the measured windows instead); and everything downstream of $\bar{\rho} \propto \varphi_{\max}^2$ — u, δt_c, Δ_{pre}, the ceiling of (21). The value: the box is the excitation design's own commanded cap, $\varphi_{\max} = 5°$. This is enough because every integral in the bound stops at τ_{end}: the channels, $\bar{\rho}$, and Δ_{pre} live on the fit window alone, where the realised tilt stays at or below $\approx 4.8°$ — the 9.4° excursion of the whole contact phase happens *beyond* the window end and never enters. (The envelope check on the exact nonlinear solution confirms the margin: $e_\omega(\tau_{\text{end}}) \leq 0.29\,E(\tau_{\text{end}})$ at every rate with the 5° $\bar{\rho}$.) Every appearance is monotone in φ, so the box only over-covers. One consequence is structural: at this box the model term no longer subsidises the noise estimate — with the old 10° box its slack was absorbing the estimation error of $\hat{n}$ on the tail runs; that duty now sits explicitly with the envelope (19b), and the model term's own slack over the realised δe_ω drops to 1.4–2.0×.

What "decays like" means — the bridge from (16) to (16a). Nothing in the window decreases with time: φ and ω both grow. The decay is the same fact read *backwards from the window end* — the claim under each brace of (16) is

$$|\dot{\rho}_{\text{channel}}(s)| \;\leq\; |\dot{\rho}_{\text{channel}}(\tau_{\text{end}})|\, e^{kC_2(s-\tau_{\text{end}})}, \qquad s \leq \tau_{\text{end}},\quad k \in \{2, 1\},$$

where the exponential factor is < 1 because $s - \tau_{\text{end}} < 0$: at earlier times the forcing sits below its end value by at least that factor. This is the form the comparison problem of Step 6 consumes — a peak at the end with a known fall-off behind it. Divide both sides by $e^{kC_2 s}$ and the claim becomes $|\dot{\rho}(s)|/e^{kC_2 s} \leq |\dot{\rho}(\tau_{\text{end}})|/e^{kC_2\tau_{\text{end}}}$ — i.e. *the ratio $|\dot{\rho}(s)|/e^{kC_2 s}$ is nondecreasing in s.* That is why (16a) is a statement about ratios. And because each channel is a *product* of trajectory factors, $\dot{\rho}/e^{kC_2 s}$ factors as a product of single-factor ratios ($\varphi\omega/e^{2C_2 s} = [\varphi e^{-C_2 s}][\omega e^{-C_2 s}]$), a product of nonnegative nondecreasing functions is nondecreasing, and the exponent k in (16) is literally a *count of trajectory factors*: two in $\varphi\,\omega$ ($k = 2$), one in each ground-effect term ($k = 1$). Two sanity checks on the direction of the conservatism. The decaying half of cosh ($e^{-C_2\tau}$) is not neglected — it sits inside the proven ratio $(1-e^{-C_2\tau})^2/2$ and only pushes the true curve *further below* the end-anchored exponential. And at $\tau = 0$ the true forcing vanishes ($\varphi = \omega = 0$) while the envelope starts at $\dot\rho_2 e^{-2x_{\text{end}}} > 0$: the envelope *over-covers* near onset — valid, equality touching only at the window end where it is anchored — at a cost of $e^{-2x_{\text{end}}}$ (2% to 0.06% of the peak on the campaign), in a region the pinned response barely feels.

Why anchor at the end rather than the onset. Four reasons, the last the deepest. A forward envelope $|\dot\rho(s)| \le |\dot\rho(0)|e^{kC_2 s}$ has nothing to hang on: the forcing *vanishes* at onset ($\varphi = \omega = 0$), so its anchor amplitude is zero; the one usable anchor is the peak — the end value, which any bound must pay anyway and which the channel sups bound a priori. Second, the tightness lives in "how little forcing there is early": the Dirichlet response has a boundary zero directly under the forcing peak, and exploiting that cancellation requires the envelope to encode the fall-off *behind* the peak — exactly what the backward rate is, and exactly what a constant (sup-only) envelope discards at a cost of 8–12×. Third, only the end-anchored coordinate $u = C_2(s-\tau_{\text{end}}) \in [-x_{\text{end}}, 0]$ makes the comparison problem forget the window length: as $x_{\text{end}} \to \infty$ the end neighbourhood is fixed and only the onset recedes, which is why the x_{end}-free constants $1/12$ and $1/(2e)$ exist at all; in an onset-anchored coordinate the peak would move with x_{end} and no such constants would emerge. Last, the instability has one clock, $1/C_2$, and everything that matters happens in the last few e-foldings before τ_{end}: *measured from the end, all windows look alike.* That statement is the coordinate version of the rate-flatness of the residual — the ramp rate only moves the onset distance x_{end}, and the end neighbourhood forgets x_{end}.

Step 5′ — the decays marked in (16) are theorems, not observations. Along the nominal trajectory, write $a = C_2 s$. By the bridge above, it suffices that each factor's $e^{C_2 s}$-ratio is nondecreasing; the first ratio of (16a) certifies the factor $\omega e^{-C_2 s}$, the second certifies $\varphi e^{-C_2 s}$, and both facts are one line each:

$$\frac{\omega_{\text{nom}}(s)}{C_1\, e^{C_2 s}} = \frac{(\cosh a - 1)}{e^a} = \frac{\left(1-e^{-a}\right)^2}{2} \nearrow, \qquad \frac{d}{da}\left[\frac{\sinh a - a}{e^a}\right] = e^{-a}\left(a - 1 + e^{-a}\right) \ge 0, \tag{16a}$$

the first read off directly, the second because $e^{-t} \ge 1-t$; the bracket in the second is $\varphi_{\text{nom}}\, C_2 e^{-C_2 s}/C_1$, so $\varphi_{\text{nom}} e^{-C_2 s}$ is nondecreasing too. Now every channel of $\dot\rho$ is a product of positive factors whose $e^{C_2 s}$-ratios are nondecreasing, and such a product obeys its decay exactly. Arm channel:

$$\sin\varphi\,\omega \;\le\; \varphi\,\omega = \left[\varphi e^{-C_2 s}\right]\left[\omega e^{-C_2 s}\right] e^{2C_2 s} \;\le\; \varphi(\tau_{\text{end}})\,\omega(\tau_{\text{end}})\, e^{2C_2(s-\tau_{\text{end}})}.$$

(The height term, had it been kept, is $(1-\cos\varphi)\,\omega \le \frac{1}{2}\varphi^2\omega$ — *three* such factors, decaying as $e^{3C_2(s-\tau_{\text{end}})}$; it is absent from (16) by the sign cancellation of Step 5, so nothing needs charging.) Ground effect: $\dot M\varphi$ carries one factor (e^{C_2} class), and $\Delta M(s)\,\omega = \dot M s\,\omega$ carries $s/\tau_{\text{end}} \le 1$ (also nondecreasing) times one, so it too is e^{C_2} class. For the *nominal* trajectory the lemma is therefore exact, with no safety factor.

Nominal to true. The true trajectory multiplies each factor by $1 + \varepsilon(s)$ with $\varepsilon = e_\omega/\omega_{\text{nom}}$ (and its integral analogue on the φ factor). Two facts keep this cheap. First, a *constant* multiplier cancels in a ratio test — numerator and anchor scale together — so the envelope is harmed only by the *variation* of ε across the window, not its size. Second, ε has an a priori ceiling in the same dimensionless number as (9a): dividing the envelope (90) by ω_{nom},

$$\varepsilon(\tau) \;=\; \frac{e_\omega}{\omega_{\text{nom}}} \;\le\; \frac{\bar\rho\sinh(C_2\tau)/(J_P C_2)}{C_1\left(\cosh(C_2\tau)-1\right)} \;=\; u\,\coth\!\left(\frac{C_2\tau}{2}\right),$$

$u = \bar\rho C_2/\dot M$ appearing for the third time. At the window end $\coth(x_{\text{end}}/2) = 1.05$–$1.4$, and inserting the *realised* $\beta/C_1 = 0.010$–0.023 in place of the envelope gives $\varepsilon(\tau_{\text{end}}) = (\beta/C_1)\coth(x_{\text{end}}/2) = 1.4$–$2.4\%$. Measured on the exact nonlinear tip-over across the seven rates, the worst excess of a true ratio over its nominal bound is 2.6%, at the slowest ramp near the window end — the formula's prediction, not a coincidence — and the 1.05 carried on $\dot\rho_2$ in (16) covers it with double margin. (A fully pre-experiment allowance is $1 + u\coth(x_{\text{end}}/2)$ — at most 1.17 at the slowest ramp, 1.02 at the fastest, the envelope $u \le 0.16$ overstating the realised β/C_1 about seven-fold. The *end-point amplitude* no longer needs any allowance at all: since the anchor correction, (16) charges $\omega_{\text{nom}}(\tau_{\text{end}}) + E(\tau_{\text{end}})$ outright, so the 1.05 covers only the interior variation.) One apparent irony resolves itself: $u\coth(C_2\tau/2)$ diverges as $\tau \to 0$, seemingly where e_ω is smallest. The divergence belongs to the envelope, not the physics: the true ε *vanishes* at onset like τ^4 (e_ω starts at sixth order — the leading forcing channel is fourth order and Duhamel integrates twice — against ω_{nom}'s second), while (90) charges a first-order $\bar\rho\sinh(C_2\tau)$ from the start. And nothing is lost where the formula fails: the relative argument is only needed where the forcing lives, in the last e-foldings, where $\coth \approx 1$; near onset the envelope's absolute slack ($e^{-2x_{\text{end}}}$ of the peak against a true forcing of zero) does the covering — which is why the measured worst excess sits near the window *end*, not the onset. The safety factor thus covers only the nominal-to-true step — the shape itself is proved.

Step 6 (optional tightening, factor 8) — use the decay of $\dot\rho$. The forcing is not constant: it peaks at the window end and decays backwards as marked in (16) and proved in Step 5′. So solve (14) once for each shape at unit peak — the same two-line comparison applies, and the solutions are still elementary. In the scaled variable $u = C_2(\tau - \tau_{\text{end}}) \in [-x_{\text{end}}, 0]$ — a *translation*, not a reflection: $du/d\tau = +C_2 > 0$, so u runs forward with τ and only the origin has moved to the window end (a reflection would be $C_2(\tau_{\text{end}} - \tau)$; the "backward" of Step 5′ is how the envelope is *described*, not the orientation of the coordinate) — for the forcing $e^{2C_2(\tau-\tau_{\text{end}})} = e^{2u}$:

$$w_2(\tau) \;=\; \frac{\dot\rho_2}{J_P C_2^2}\cdot\frac{1}{3}\left[\frac{\sinh(u+x_{\text{end}})}{\sinh x_{\text{end}}} + e^{-2x_{\text{end}}}\frac{\sinh(-u)}{\sinh x_{\text{end}}} - e^{2u}\right], \qquad M_2(x_{\text{end}}) \;\triangleq\; \max_{u\in[-x_{\text{end}},0]}\frac{1}{3}\big[\,\cdots\big], \tag{17a}$$

and for the forcing $e^{C_2(\tau-\tau_{\text{end}})} = e^{u}$, which is *resonant* — it decays at the operator's own rate, so the particular solution carries a factor of u:

$$w_1(\tau) \;=\; \frac{\dot\rho_1}{J_P C_2^2}\left[\frac{x_{\text{end}}}{2}\,\frac{\sinh(u+x_{\text{end}})}{\sinh x_{\text{end}}} - \frac{(u+x_{\text{end}})\,e^{u}}{2}\right], \qquad M_1(x_{\text{end}}) \;\triangleq\; \max_{u\in[-x_{\text{end}},0]}\big[\,\cdots\big]. \tag{17b}$$

No time reversal hides in (17a): substituting $u = C_2\tau - x_{\text{end}}$ back gives, in original time,

$$w_2(\tau) = \frac{\dot\rho_2}{J_P C_2^2}\cdot\frac{1}{3}\left[\frac{\sinh(C_2\tau)}{\sinh x_{\text{end}}} + e^{-2x_{\text{end}}}\frac{\sinh(C_2(\tau_{\text{end}}-\tau))}{\sinh x_{\text{end}}} - e^{2C_2(\tau-\tau_{\text{end}})}\right],$$

which vanishes at $\tau = 0$ $(0 + e^{-2x_{\text{end}}} - e^{-2x_{\text{end}}})$ and at $\tau = \tau_{\text{end}}$ $(1 + 0 - 1)$, with the forcing term growing forward in τ to its peak at the end — exactly the orientation of the physical window. (Backward time $v = \tau_{\text{end}} - \tau$ would give the mirror parametrisation $w(\tau_{\text{end}} - v)$ with forcing $e^{-2C_2 v}$ — the same function, described from the other end.)

Where the two forms come from — construction, not guess. In the scaled variable the comparison problem reads $w'' - w = -\kappa\, g(u)$, $w(-x_{\text{end}}) = w(0) = 0$, with $\kappa = \dot\rho_i/(J_P C_2^2)$, one shape g at a time. (One label trap, worth naming: these are the *same* two conditions as $w(\tau{=}0) = w(\tau_{\text{end}}) = 0$, relabelled — since the origin of u sits at the window end, the token "0" names the *onset* in τ-labels but the *window end* in u-labels: $u = -x_{\text{end}}$ is $\tau = 0$, and $u = 0$ is $\tau = \tau_{\text{end}}$. Two physical pins, onset and end, in either language.) Three moves each. *Particular solution:* for $g = e^{2u}$ try Ae^{2u}: $(2^2-1)A = -\kappa$, so $A = -\kappa/3$ — the $1/3$ is the distance to resonance. For $g = e^{u}$ the same try fails, $(1^2 - 1)A = 0$: the forcing decays at the operator's own rate, so carry a factor u, $w_p = -(\kappa/2)\,ue^{u}$, since $(ue^u)'' - ue^u = 2e^u$ — this extra u is why $M_1 > M_2$. (Where the u comes from: the non-resonant recipe gives $e^{\lambda u}/(\lambda^2-1)$, which blows up as $\lambda \to 1$; subtracting the homogeneous e^u first, $(e^{\lambda u} - e^u)/(\lambda-1)\cdot 1/(\lambda+1) \to \partial_\lambda e^{\lambda u}\big|_{\lambda=1}\cdot\frac{1}{2} = \frac{1}{2}\,ue^u$ — the u is the derivative left behind when two merging exponentials collide, and the $\frac{1}{2}$ is $1/(\lambda+1)$ at $\lambda = 1$: the secular term of a resonantly driven system, in hyperbolic dress.) *Homogeneous corrections in the endpoint basis:* $\sinh(u + x_{\text{end}})$ vanishes at $u = -x_{\text{end}}$ and $\sinh(-u)$ vanishes at $u = 0$, so each boundary condition sets one coefficient without disturbing the other. (Both are legitimate homogeneous solutions — the operator has constant coefficients, so translating a solution gives a solution, and $\sinh(u + x_{\text{end}}) = \cosh x_{\text{end}}\sinh u + \sinh x_{\text{end}}\cosh u$ is just a combination of the familiar pair; the two are independent since the coefficient determinant is $\sinh x_{\text{end}} \neq 0$. Solving in the $\{\sinh u, \cosh u\}$ basis instead gives a coupled 2×2 system and the same answer with more algebra — the endpoint basis is the choice that makes the boundary conditions decouple, the same pattern that builds Green's functions.) Then: $u = 0$ gives $a\sinh x_{\text{end}} = \kappa/3$, $u = -x_{\text{end}}$ gives $b\sinh x_{\text{end}} = (\kappa/3)e^{-2x_{\text{end}}}$ — that is (17a). For e^u one more coefficient asks to be determined: the resonant particular solution $-(\kappa/2)ue^u$ can carry any multiple of e^u on top, since the operator annihilates it — that *is* the resonance — so its coefficient is a homogeneous degree of freedom, fixed by the boundary, not the equation. The condition at $u = -x_{\text{end}}$ (where the $\sinh(u + x_{\text{end}})$ term is blind) sets it to $-(\kappa/2)x_{\text{end}}$, bundling into $-(\kappa/2)(u + x_{\text{end}})e^u$ with the onset zero visible in the factor; the second endpoint coefficient then comes out zero, because everything else already vanishes there (unlike the e^{2u} case, whose particular solution does not). The remaining condition at $u = 0$ sets the sinh coefficient — that is (17b). *Verification by substitution:* the sinh's are annihilated by the operator, the exponentials return the forcing, and both brackets vanish at both ends — the two boundary zeros of (14). Their maxima barely depend on x_{end} and tend to simple constants,

$$M_2(x_{\text{end}}) \to \tfrac{1}{12}, \qquad M_1(x_{\text{end}}) \to \tfrac{1}{2e}, \qquad \text{so} \qquad |\delta e_\omega| \;\le\; \frac{M_2\dot\rho_2 + M_1\dot\rho_1}{W z_{CoM}}. \tag{17}$$

The limits are one-line maximisations: as $x_{\text{end}} \to \infty$, $\sinh(u + x_{\text{end}})/\sinh x_{\text{end}} \to e^u$ and the $e^{-2x_{\text{end}}}$ term dies, so the bracket of (17a) tends to $(e^u - e^{2u})/3 = (v - v^2)/3$ with $v = e^u \in (0, 1]$, maximised at $v = \frac{1}{2}$

to $1/12$ ($\ln 2$ e-foldings before the window end); the bracket of (17b) tends to $-(u/2)e^u$, maximised at $u = -1$ to $1/(2e)$ (one e-folding before the end). The limits are moreover *proven ceilings* for every finite x_{end}, in two lines. The addition theorem, divided through, gives the exact identity $\sinh(u + x_{\text{end}})/\sinh x_{\text{end}} = e^u + e^{-x_{\text{end}}} \sinh(u)/\sinh x_{\text{end}}$; inserting it splits each finite-x_{end} bracket into its limit form plus a correction,

$$B_2(u; x_{\text{end}}) = \left(e^u - e^{2u}\right) + \frac{\sinh u}{\sinh x_{\text{end}}}\left(e^{-x_{\text{end}}} - e^{-2x_{\text{end}}}\right), \qquad B_1(u; x_{\text{end}}) = -\frac{u}{2}\, e^u + \frac{x_{\text{end}}\, e^{-x_{\text{end}}}}{2 \sinh x_{\text{end}}} \sinh u,$$

and on $u \in [-x_{\text{end}}, 0]$ the correction is ≤ 0 in both cases — a positive coefficient times $\sinh u \le 0$. Each bracket is nonnegative (the Dirichlet response to a nonnegative forcing, by (14) itself), so its maximum sits below the limit's maximum: $M_2(x_{\text{end}}) \le 1/12$ and $M_1(x_{\text{end}}) \le 1/(2e)$ for every window. The correction term is precisely the homogeneous piece that enforces the onset pin at $u = -x_{\text{end}}$, which the infinite-window problem lacks — an extra pin on a positive response can only pull it down: a shorter window gives the forcing less room. (Over the campaign range $x_{\text{end}} = 1.9$–3.7: $M_2 = 0.074$–0.083, $M_1 = 0.141$–0.181; the script evaluates (17a)–(17b) exactly.) The interpretation is the same as the manuscript's reduction factors $1/7$ and $1/5$: the forcing is largest exactly where the pinned response is smallest — at the window end the boundary zero sits right under the peak of the load.

A one-line version, in the spirit of (85). The limits are also *ceilings*: $M_2(x_{\text{end}}) \le 1/12$ and $M_1(x_{\text{end}}) \le 1/(2e)$ for every finite window — proved after (17) by the exact bracket split, whose correction term is negative on the whole interval — so (17) may be quoted x_{end}-free,

$$|\delta e_\omega| \;\le\; \frac{\dot\rho_2/12 \;+\; \dot\rho_1/(2e)}{W z_{CoM}},$$

with no M curves, no window length, and no C_2 beyond what sits inside $\dot\rho_{1,2}$ — exactly as (85) trades the exact remainder for $\frac{1}{2}G''(0)\varphi^2$. Rerun on the 140 runs, the constant version costs at most 5% on the worst run (0.1–2% in the per-rate mean, largest at the fastest ramps where x_{end} is smallest), keeps 140/140 coverage, and the worst-run used fraction rises only to 0.67. Note where the simplification is *not* available: the channel amplitudes $\dot\rho_{1,2}$ are already the $\varphi^* = 0$ worst-case Taylor coefficients of (85) — nothing further to drop there — and the time *shape* behind M_2, M_1 (the forcing concentrated at the window end as $e^{2C_2(s-\tau_{\text{end}})}$) cannot be dropped at all: against a constant-in-time forcing the comparison factor is $1 - \operatorname{sech}(x_{\text{end}}/2) = 0.4$–$0.9$, a factor 5–10 above $1/12$, which is the loose bound this document exists to escape. The Taylor step simplifies the *amplitude*; the tightness lives in the *shape*.

6 The pre-onset segment

Where the segment comes from. By (9), $\delta t_c < 0$: the witness branch starts $|\delta t_c|$ *before* the true onset. On the segment $\tau \in [\delta t_c, 0]$ — length $|\delta t_c|$ — the record is still at its baseline, but the branch has already been running for $u = \tau - \delta t_c \in [0, |\delta t_c|]$ since its own onset, so the mismatch there is the branch itself:

$$\text{mismatch}(u) \;=\; C_1\big(\cosh(C_2 u) - 1\big), \qquad u \in [0,\, |\delta t_c|]. \tag{18a}$$

The residual sum runs over this segment too, so the witness's cost in (10) gains its square. Normalising by the main window length τ_{end} (conservative: the extended window is longer, so dividing by the shorter τ_{end} can only enlarge the term),

$$\Delta^2_{\text{pre}} \;=\; \frac{1}{\tau_{\text{end}}} \int_0^{|\delta t_c|} \left[C_1\big(\cosh(C_2 u) - 1\big)\right]^2 du. \tag{18b}$$

The segment's *noise* adds no separate term: $\hat n$ is an RMS density, unchanged by a slightly longer window.

The integral is elementary. Expand the square with $\cosh^2 a = \frac{1}{2}(1 + \cosh 2a)$:

$$\big(\cosh a - 1\big)^2 = \cosh^2 a - 2\cosh a + 1 = \tfrac{3}{2} + \tfrac{1}{2}\cosh(2a) - 2\cosh a, \tag{18c}$$

and integrate term by term with $a = C_2 u$ ($\frac{3}{2} \to \frac{3u}{2}$; $\frac{1}{2}\cosh(2C_2 u) \to \sinh(2C_2 u)/(4C_2)$; $-2\cosh(C_2 u) \to -2\sinh(C_2 u)/C_2$):

$$\Delta^2_{\text{pre}} \;=\; \frac{C_1^2}{\tau_{\text{end}}} \left[\frac{3u}{2} + \frac{\sinh(2C_2 u)}{4C_2} - \frac{2\sinh(C_2 u)}{C_2}\right]_0^{|\delta t_c|}. \tag{18}$$

Check, and why the term is small. Expanding the bracket in powers of u, the u and u^3 orders cancel identically and the first surviving order is

$$\int_0^{|\delta t_c|} \big(\cosh(C_2 u) - 1\big)^2 du \;\approx\; \frac{C_2^4\,|\delta t_c|^5}{20},$$

fifth order in the shift, because the branch leaves the baseline only quadratically ($\cosh -1 \approx C_2^2 u^2/2$, squared $\to u^4$). Bounding instead by (segment sup)×(length) would give $|\delta t_c|^5/4$ — the exact integral is a factor $\sqrt{5} \approx 2.2$ tighter. (The branch amplitude is $R \le C_1$; using C_1 in (18) errs on the high side.) Evaluated with the a priori $|\delta t_c|$ from (9) and $e_\omega(\tau_{\text{end}}) \le E(\tau_{\text{end}})$: $\Delta_{\text{pre}} = 0.001°$/s at $\dot{M} = 0.10$ and below 0.001 beyond — with the 5° box the a priori shift is 23–28 ms at the slowest ramp and the fifth-order integral makes the term vanish.

The baseline constant $R - C_1$ — both bookkeepings. One witness must serve every sample the minimiser's cost touches, and its baseline slot holds a single number, so the constant $R - C_1$ of (7) must land somewhere. There are two consistent choices. *(a)* Take (3) literally, $C = b$: the flat pre-onset samples then cost nothing beyond noise, (18a) holds as written (amplitude $R \le C_1$), and the post-onset residual gains the constant, $r = \delta e_\omega + n + (C_1 - R)$. *(b)* Take $C = b + (R - C_1)$: the post-onset identity is exactly (5), $r = \delta e_\omega + n$, and the constant moves to the flat stretch, where the length fraction of that stretch in the window dilutes it further. Either way the price is the same number,

$$C_1 - R \;=\; C_1\Big(1 - \sqrt{1 - (\beta/C_1)^2}\Big) \;\le\; \tfrac{1}{2}\, C_1\,(\beta/C_1)^2,$$

second order in β/C_1. Realised ($\beta/C_1 \le 0.023$) it is below 0.004°/s on every run — two orders under $\hat{n}$ and below even Δ_{pre} — which is why (10) and (20) carry it silently; the pre-onset median cannot even distinguish b from $b + (R - C_1)$ at that size. Even at the a priori envelope ceiling it stays invisible: $u \le 0.16$ gives $C_1(1 - \sqrt{1 - u^2}) \le 0.014°$/s at the slowest ramp — below Δ_{pre}'s old level — so the silent treatment needs no a posteriori excuse.

7 The noise term

Above 5 Hz the model can produce nothing, so n_{hi} is measured from the record; the full band is estimated through the quiet-window shape ratio,

$$\hat{n} \;=\; \text{RMS}(n_{\text{hi}})\sqrt{1 + (s_{\text{med}}\,\kappa_q)^2}, \tag{19}$$

with $s_{\text{med}} = 1.60$ the campaign-median in-window enhancement and κ_q the run's own quiet ratio (campaign median 0.37 where no quiet stretch exists).

Reading (19) piece by piece. The cap needs the *total* in-window disturbance RMS(n), and no part of the record hands it over directly: below ∼1 Hz the residual is model-contaminated, and the pre-onset floor underestimates the contact-borne in-window disturbance (next paragraph). So (19) assembles it in three steps. *Anchor:* above 5 Hz the branch has no spectral content, so the high-passed residual n_{hi} is pure disturbance, measured in the window itself — RMS(n_{hi}) is the one number taken from the run on trust. Note what is *not* needed here: a ground-truth curve. The separation is by frequency, not by subtracting a truth — every rigid-body term in the record (ω_{nom}, e_ω, b) is band-limited below a few hertz because the dynamics run at $C_2/2\pi < 1$ Hz, so in the high band $r_{\text{hi}} = y_{\text{hi}} - \hat{\omega}_{\text{hi}} = n_{\text{hi}}$ identically: nothing else has a component left there. n_{hi} is *defined* by the band, not reconstructed from a reference. *Shape:* the disturbance's low-band-to-high-band ratio is measured where no model exists at all, on quiet stretches of the same record — that is κ_q. *Enhancement:* the in-window disturbance is redder than quiet-time noise (gear-edge pivoting pumps the low band), so the quiet ratio is scaled by the campaign-median in-window enhancement $s_{\text{med}} = 1.60$, making the estimated in-window low/high ratio $s_{\text{med}}\kappa_q$. The two bands are disjoint in frequency, hence orthogonal, so their RMS values add in quadrature: $\text{RMS}(n)^2 = \text{RMS}(n_{\text{hi}})^2\big(1 + (s_{\text{med}}\kappa_q)^2\big)$ — which is (19).

The envelope form that the cap uses. (19) is an *estimate* — s_{med} is a campaign median, so on roughly half the runs the true disturbance exceeds $\hat{n}$. An estimate is the right tool for interpretation (Sec. 7) but not for a guarantee, so the cap (20) carries the envelope form instead:

$$\hat{n}_b \;=\; \text{RMS}(n_{\text{hi}})\sqrt{1 + \kappa_{\text{sup}}^2}, \qquad \kappa_{\text{sup}} = \max_{\text{runs}} \kappa_{\text{imp}} = 1.69, \tag{19b}$$

the campaign maximum of the in-window ratio itself — the *direct* pooled maximum of the quantity the bound needs. (An intermediate version bounded it indirectly, through the quiet ratios, as $s_{\text{med}}\kappa_b = 2.09$ with

$\kappa_b = \max \kappa_q$; the direct maximum is 24% tighter with the same status.) The multiplier is $\sqrt{1 + 1.69^2} = 1.96$ on $\mathrm{RMS}(n_{\mathrm{hi}})$: $\hat{n}_b$ is anchored at the largest measured ratio — it upper-bounds every observed run's disturbance, with equality on the one run that sets the maximum, where the model term supplies the margin. The maximum is not an isolated freak: five runs sit above $0.8\,\kappa_{\mathrm{sup}}$. This is what frees the model term from double duty: with (19) in the cap, the model term's slack was absorbing the noise-estimation error of the tail runs; with (19b) each term guards exactly its own object — the model term δe_ω, the noise envelope the disturbance.

Why the pooled maximum and not a pooled mean+3σ. The natural statistical alternative pools the 140 in-window ratios and uses their mean+3σ: it would tighten the multiplier from 1.96 only to 1.88 — 4% — and it is not sound here, because the data refuses the Gaussian tail the 3σ logic assumes: the pooled κ_{imp} distribution reaches its maximum at 3.3σ above its mean, so mean+$3\sigma = 1.59$ sits *below* the observed maximum 1.69 — the statistical noise term alone would leave the worst run to the model term, quietly reintroducing the double duty (19b) exists to remove. The pooled *maximum* is itself a campaign statistic, needs no distributional assumption, and keeps the guarantee unconditional over everything observed — at a price of 4% against an assumption the data visibly violates.

Implementation, and the picture (Fig. 2). The split is a brick-wall FFT partition: remove the mean, zero the bins above f_c, invert — so the two bands are *exactly* orthogonal and the quadrature sum above is Parseval-clean. (Orthogonal in the plain vector sense: $\sum_i r_{\mathrm{lo},i}\, r_{\mathrm{hi},i} = 0$. The N samples are a vector in $\mathbb{R}^N$, the FFT rewrites it in an orthogonal basis of sinusoids — one basis direction per bin — and the brick wall hands each direction *wholly* to one band, so the two parts are built from disjoint coordinates and their dot product vanishes term by term; then $\|r\|^2 = \|r_{\mathrm{lo}}\|^2 + \|r_{\mathrm{hi}}\|^2$ with no cross term. Checked on all 140 runs: the normalised inner product and the Pythagoras error are 6×10^{-16}, machine precision. A gradual filter would share the transition-band bins between the two parts: the same data under a Butterworth-like mask gives an inner product cosine of 0.47 and a 19% Pythagoras error — the brick wall is what buys the equals sign.) κ_q applies the same partition to a quiet stretch *of the same length as the window*, taken from the same record before the excitation; s_{med} is the campaign median of $\kappa_{\mathrm{imp}}/\kappa_q$, where κ_{imp} is the low/high ratio the window itself implies — and its definition is nothing but Pythagoras inverted. The partition is orthogonal, so $\mathrm{RMS}(r)^2 = \mathrm{RMS}(r_{\mathrm{lo}})^2 + \mathrm{RMS}(r_{\mathrm{hi}})^2$ exactly; both $\mathrm{RMS}(r)$ and $\mathrm{RMS}(r_{\mathrm{hi}}) = \mathrm{RMS}(n_{\mathrm{hi}})$ are already in hand, so no second measurement is needed:

$$\kappa_{\mathrm{imp}} \;=\; \frac{\mathrm{RMS}(r_{\mathrm{lo}})}{\mathrm{RMS}(r_{\mathrm{hi}})} \;=\; \frac{\sqrt{\mathrm{RMS}(r)^2 - \mathrm{RMS}(n_{\mathrm{hi}})^2}}{\mathrm{RMS}(n_{\mathrm{hi}})}.$$

One honest caveat, in the safe direction: r_{lo} contains the unabsorbed model remainder δe_ω as well as low-band disturbance, so κ_{imp} — and with it s_{med} — errs high: the enhancement conservatively charges any model leftover to the noise term's low band too, and the cap can only grow for it. Two clarifications that head off natural objections. *No circularity — but stated carefully:* n_{hi} is computed from the minimiser residual, the same residual the cap bounds. It is not *fit-free*: splitting the raw record fails (the finite window's truncated branch leaks into the high bins themselves, 4–14× the disturbance, and only subtracting a curve that carries the same leakage cancels it). It is *fit-robust*, which is what breaks the circle: the high-band RMS is insensitive to *which* good smooth curve is subtracted — swapping the C_2-pinned two-parameter refit for the fully free fit changes it by 1.4% at the median (16% worst) — so the anchor does not reward the minimiser for fitting well; any member of the family delivers the same number.

Why not the textbook noise estimators. The two standard fit-free estimators — the second-difference formula $\sigma^2 = \sum (y_{i+1} - 2y_i + y_{i-1})^2/6N$ and the finest-scale wavelet MAD of Donoho–Johnstone — run on the raw record without any fit, and both *under*-read this disturbance: 0.47× and 0.65× of $\mathrm{RMS}(n_{\mathrm{hi}})$ at the campaign median. The reasons are structural, not tuning. Both anchor at the very top of the band (the second difference weights the spectrum by $|2\sin(\pi f/f_s)|^4$, concentrated near Nyquist; the finest wavelet scale is the top octave), and the contact-borne disturbance is *coloured* — falling above a few hertz — so the top of the band is its quietest part; and the MAD variant additionally discounts the bursts that carry much of a contact disturbance's energy. The under-read is systematic, not random: the ratios are nearly constant across the seven rates (0.42–0.48 and 0.57–0.71). Swapped into (19b) as the anchor, the second-difference cap fails on 59 of the 140 runs (worst run at 2.16× its cap) and the MAD cap on 13 (worst 1.86×). (19)'s anchor is the same idea made band-honest: integrate the whole model-free band instead of sampling its quietest corner.

A model-curve-free anchor exists, and it confirms the deployed one. For a reader uneasy that the method's own curve appears in the anchor at all, three generic estimators handle the truncation leakage without ever seeing the cosh family (`analysis/neutral_anchor.py`): a degree-4 Legendre detrend before the split, a DCT brick-wall split (the even extension removes the periodicity jump that defeats the raw DFT split), and a Savitzky–Golay residual. Against the deployed anchor their medians are 0.97, 1.19 and 0.97 — independent confirmation that the cosh subtraction is not flattering the method — but each has runs where generic

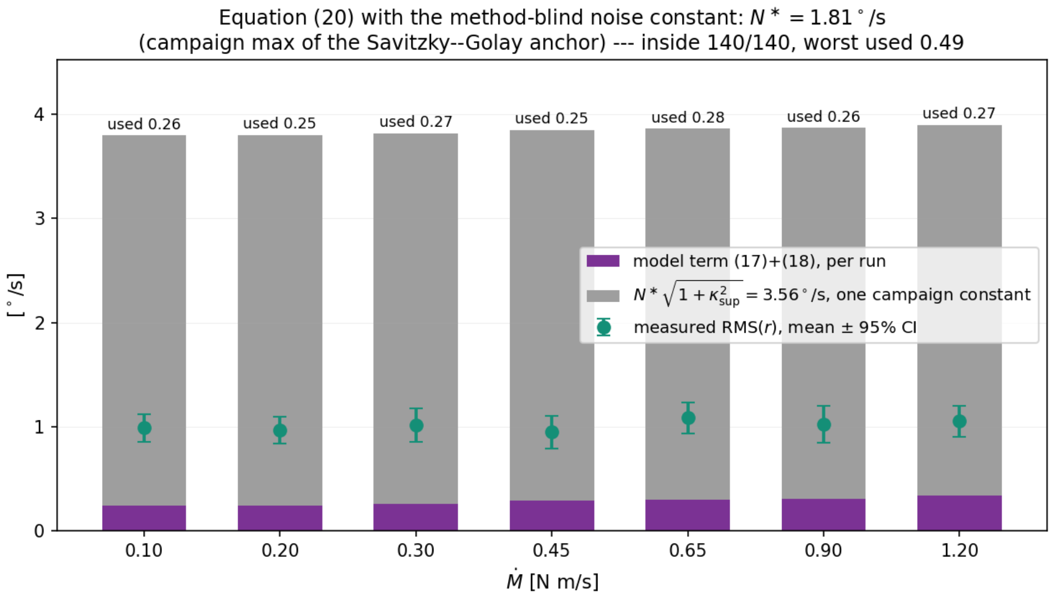

Figure 1: Equation (20) with the fully method-blind noise constant: the campaign maximum of the Savitzky–Golay anchor, $N^* = 1.81°/\mathrm{s}$, times $\sqrt{1+\kappa_{\mathrm{sup}}^2}$ — one grey number for every run, no model curve anywhere in the noise side. Purple is the per-run model term, green the measured residual with 95% intervals. All 140 runs stay inside at used 0.25–0.28: the price of full method independence is a factor ~2 against the deployed per-run anchor.

smoothing absorbs genuine disturbance (on one run all three under-read by 38%), so singly they cover 139/140. The per-run maximum of the three, inflated 1.10×, is a valid fully model-free cap: 140/140, worst used 0.94, at a mean used of 0.39 against the deployed 0.50–0.52. The deployed anchor stays — tighter, fit-robust, and now independently corroborated. The simplest member of this family is also worth stating: take the campaign *maximum* of the Savitzky–Golay anchor as a single constant, $N^* = 1.81°/\mathrm{s}$, and the noise term becomes one method-blind number, $N^*\sqrt{1+\kappa_{\mathrm{sup}}^2} = 3.6°/\mathrm{s}$ for every run — the cap still holds 140/140 (worst used 0.49, mean 0.26). The three anchors form a ladder: per-run with the branch subtracted (used 0.50–0.52), per-run and model-blind (0.39), one model-blind campaign constant (0.26) — each step buys independence from the method at a factor in slack, and the reader may stop wherever their scepticism is satisfied. Fig. 1 draws the constant-anchor rung against the campaign.

No quiet data in the amplitude: the quiet stretch enters only through the dimensionless shape ratio κ_q, never through the level — deliberately, since the in-window contact-borne disturbance runs 2 to 50 times the quiet floor; amplitude from the window, shape from quiet, and s_{med} to correct the shape for the window. Fig. 2(a) shows the raw material itself — the discrete FFT bins of one run's residual and quiet stretch, $\Delta f = 1/T$ apart, split at f_c with no taper and no padding: these bins are the exact objects the numbers of (19) sum over. Fig. 2(b) draws the same run's smoothed spectra: the fitted branch tracks the record up to a few hertz and then plunges — above the split it can produce nothing, so there the residual coincides with the record and is disturbance by construction; the quiet stretch sits below it with its own shape. Fig. 2(c) shows the 113 pairs $(\kappa_q, \kappa_{\mathrm{imp}})$ behind s_{med}.

The cutoff is not delicate, and $k\sigma$ is what (19) already is. Structurally (19) reads $k \cdot \sigma$ with $\sigma = \mathrm{RMS}(n_{\mathrm{hi}})$ — measured *in the window*, in a band the model cannot contaminate — and $k = \sqrt{1+(s_{\mathrm{med}}\kappa_q)^2} \approx 1.6$–$1.9$. The split frequency only defines that band, and any choice works that sits well above the model's spectral content $C_2/2\pi = 0.6$–0.9 Hz and leaves spectral bins on both sides (a *bin* is one discrete frequency of the window's FFT, $f_k = k/T$, spaced $1/T \approx 1.4$–2.5 Hz apart here — the shortest windows have only two or three below 5 Hz, hence $f_c \gtrsim 2/(NT_s) \approx 3$ Hz): recomputing the entire check at $f_c = 2$, 3 and 5 Hz gives identical results (140/140 inside, mean used 0.57–0.58), and only past 8 Hz does the shrinking high band make the estimate noisy (133/140 at 8, 129/140 at 10). The invariance is expected: $\hat{n}$ estimates the total $\mathrm{RMS}(n)$, which does not depend on where the record is split. What does *not* work is the simpler $k\sigma$ with σ from the pre-onset segment: the in-window disturbance is contact-borne — pivoting on one gear edge, not gyro noise — and runs 2 to 50 times the pre-onset floor with no usable floor at all on 25 of the 140 runs, so the one campaign constant that covers the spread, $k = 49$, collapses the used fraction from 0.58 to 0.16.

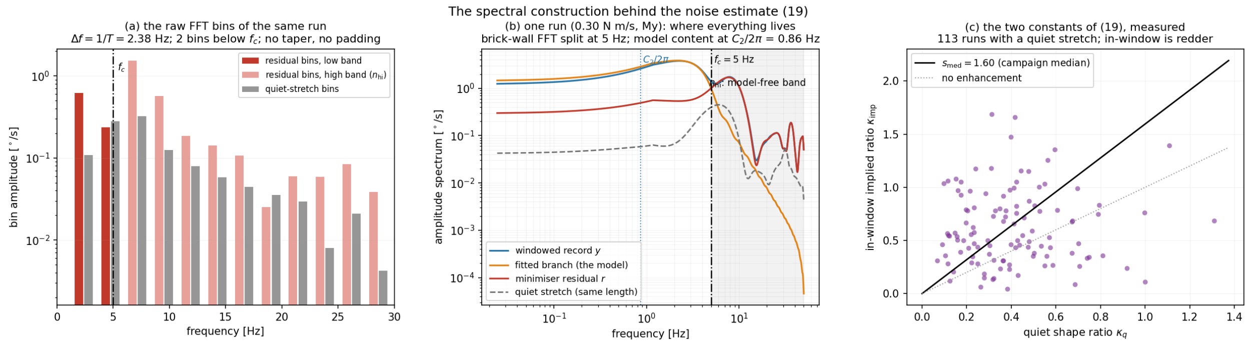

Figure 2: The spectral construction of (19). *(a)* The raw FFT bins of one run's residual and quiet stretch — one bar per discrete frequency $f_k = k/T$, spacing $\Delta f = 1/T$, split at f_c with no taper or padding: exactly the objects the numbers of (19) sum over, low band (dark) versus the anchor n_{hi} (light). *(b)* The same run's smoothed amplitude spectra: the fitted branch (the model) collapses above a few hertz, so past the 5 Hz split the residual coincides with the record — pure disturbance; the quiet stretch of the same length gives the shape ratio κ_q. These display spectra are Hann-tapered and zero-padded *for display only* — the numbers in (19) use the raw bins of panel (a), where the branch has been subtracted and edge leakage cancels. *(c)* The in-window implied ratio κ_{imp} (Pythagoras inverted on the window's own residual) against the quiet ratio κ_q over the 113 runs with a usable quiet stretch; the in-window disturbance is redder, by the campaign-median factor $s_{\text{med}} = 1.60$.

8 The check on all 140 runs

$$\boxed{\mathrm{RMS}(r) \;\le\; \underbrace{\frac{M_2\dot\rho_2 + M_1\dot\rho_1}{W z_{CoM}} + \Delta_{\text{pre}}}_{\text{model, before the experiment}} \;+\; \hat n_b} \tag{20}$$

The three terms have three different statuses, worth naming. The first is a *sup* bound standing in for an RMS: (17) caps $|\delta e_\omega(\tau)|$ at every point, and $\mathrm{RMS}(\delta e_\omega) \le \sup|\delta e_\omega|$ always — a conservative substitution, not an approximation. The second is an RMS *by definition*, (18b). The third upper-bounds $\mathrm{RMS}(n)$ by the envelope (19b) — the estimate (19) serves the interpretation, the envelope serves the guarantee. They combine by the triangle inequality of the L^2 norm, $\mathrm{RMS}(A+B) \le \mathrm{RMS}(A) + \mathrm{RMS}(B)$; the first two live on disjoint segments and are therefore orthogonal, so quadrature ($\sqrt{\text{model}^2 + \Delta_{\text{pre}}^2}$) would be admissible and slightly tighter — linear addition is kept as the conservative, simpler form, and with $\Delta_{\text{pre}} \le 0.036°/\text{s}$ the difference is invisible.

The RMS-of-w variant: how much the sup substitution costs. The genuinely available tightening sits in the first term. Since (17) caps $|\delta e_\omega(\tau)|$ pointwise by the comparison solution $w(\tau)$, one may use $\mathrm{RMS}(\delta e_\omega) \le \mathrm{RMS}(w)$ instead of $\sup w$ — and because w is concentrated in the last e-foldings, its RMS over the window is genuinely smaller, with a $1/\sqrt{x_{\text{end}}}$ dilution the sup ignores: asymptotically $\mathrm{RMS}(B_2) \to \sqrt{1/(108\,x_{\text{end}})}$ and $\mathrm{RMS}(B_1) \to \sqrt{1/(16\,x_{\text{end}})}$ (the integrals $\int_{-\infty}^{0} B_{2,\infty}^2\,du = 1/108$ and $\int u^2 e^{2u}/4\,du = 1/16$). Evaluated exactly on the campaign (`analysis/rms_w_variant.py`): the model term tightens by 1.43× at the fastest ramp to 1.69× at the slowest (where x_{end} is largest) — but with the envelope (19b) now dominating the cap, the total barely moves: mean used 0.50–0.52 against 0.53–0.56, coverage 140/140 either way, worst run 0.77 against 0.83. With each term guarding its own object, the norm choice inside the sub-dominant model term is a matter of taste; the sup form is kept for its simpler statement.

$\dot M$ N m/s	model (17)+(18) °/s	of it (18) °/s	$\hat n_b$ °/s	cap (20) °/s	measured °/s	used
0.10	0.238	0.001	1.750	1.99	0.987	0.50
0.20	0.243	0.000	1.686	1.93	0.964	0.50
0.30	0.257	0.000	1.780	2.04	1.014	0.50
0.45	0.289	0.000	1.610	1.90	0.947	0.50
0.65	0.302	0.000	1.767	2.07	1.084	0.52
0.90	0.308	0.000	1.654	1.96	1.023	0.52
1.20	0.339	0.000	1.683	2.02	1.052	0.52

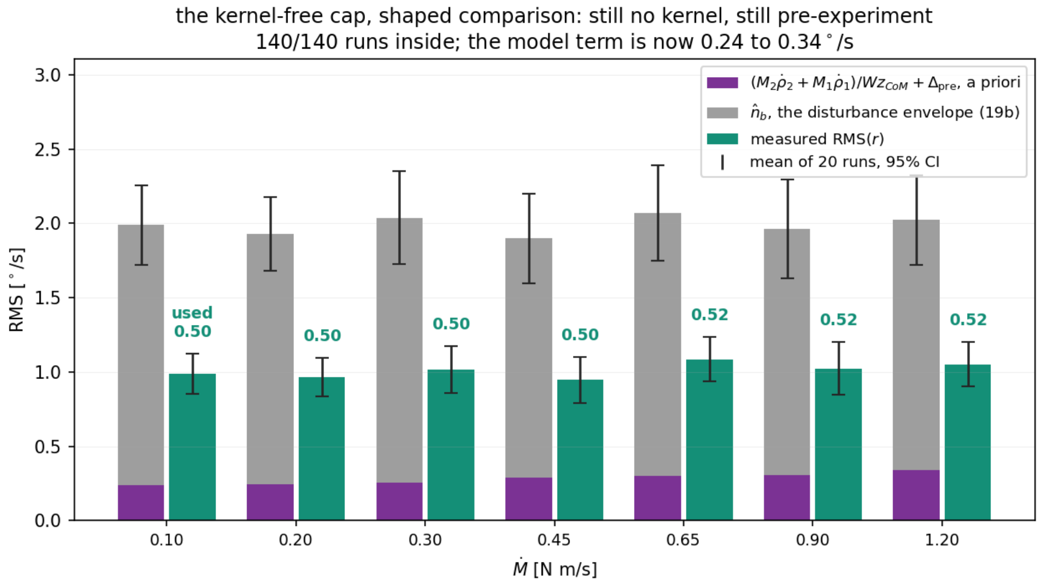

Figure 3: Equation (20) against the campaign: means over the twenty runs at each rate, 95% t intervals. Purple is the model term of (17)+(18), grey the noise envelope (19b), green the measured residual. All 140 runs are inside; the used fraction is 0.50–0.52 and the cap is flat, like the residual.

Inside on 140/140 runs; per-run used fraction p10 0.44, median 0.49, worst 0.77. (That the $\hat{n}_b$ column exceeds the measured residual is by construction, twice over: $\hat{n}_b$ is the *envelope* (19b), a bound on the disturbance rather than its level — the level estimate (19) sits at the residual — and even the true disturbance generically exceeds the residual slightly, because the fit absorbs the noise component parallel to the family: $\mathbb{E}[\mathrm{RMS}(r)^2] = \sigma^2(1 - p/N)$, the projection $\|(I-P)n\| \leq \|n\|$ of the appendix.)

The table's most striking feature — measured ≈ $\hat{n}$ — and what it does and does not prove. The measured residual sits at the noise estimate (campaign median of $\mathrm{RMS}(r)/\hat{n}$: 0.976, IQR 0.89–1.09): the residual is statistically indistinguishable from the disturbance floor, i.e. the fit absorbed the model error — the realised δe_ω of 0.16–0.17°/s is invisible under a 1°/s floor ($\sqrt{1^2 + 0.17^2} = 1.014$). One part of this is by construction and must not be cited as evidence: on the 113 runs with their own quiet stretch the median of $\mathrm{RMS}(r)/\hat{n}$ is exactly 1.000 with 50% above — $s_{\rm med}$ *is* the median of $\kappa_{\rm imp}/\kappa_q$, so the median run's $\hat{n}$ is calibrated to its own residual. What is *not* by construction, and is the actual evidence: the spread stays narrow (the calibration pins the median, not the IQR), the ratio is flat in $\dot{M}$, and the upper tail — runs up to $1.76\,\hat{n}$ — is covered by the noise envelope (19b), anchored at the largest measured ratio $\kappa_{\rm sup}$, with the model term adding the guaranteed δe_ω headroom on top. The 140/140 coverage therefore rests on the envelope and the model term, not on the calibration.

(What "measured" is, precisely: the onset is placed by the paper's PNLS estimator, and the residual is then the least-squares refit of amplitude and baseline on that window with C_2 pinned to the calibrated value. The full minimiser — onset and exponent free, the family the proof actually covers — fits at least as well, so its residual is ≤ the number in the table: coverage is claimed against the harder residual.) On the exact nonlinear solution the realised δe_ω is 0.16–0.17°/s; the model term sits only 1.4×–2.0× above it — with the box at the excitation cap and the anchor at the true rate, most of the old slack is gone, and what remains is the product-of-sups and the shape transfer.

9 The two answers this gives

Why the residual is flat in $\dot{M}$ — and why the model term of the cap is not. The table shows two different rate behaviours, and they must be kept apart. The *residual* is flat because it is noise-floored: the realised unabsorbed remainder is 0.16–0.17°/s at *every* rate, invisible under the ≈ 1°/s disturbance floor, so the theory's observable prediction is $\mathrm{RMS}(r) \approx \hat{n}$, flat — and the data agrees: $\mathrm{RMS}(r) = 0.95$–1.08°/s across a twelvefold change in $\dot{M}$ (correlation $r = +0.09$, $p = 0.31$). The *model term* of (20), by contrast, grows ×1.4 across the sweep, from one identifiable factor: the true end rate genuinely grows (faster ramps tip faster through the same tilt — the nominal part alone grows ×1.6), and the bound charges the box tilt $\varphi_{\max}$ *at* that end rate — a product of sups. The true trajectory never realises both extremes together and its remainder stays flat, so the bound's slack widens, from ×1.4 at the slowest ramp to ×2.0 at the fastest — while the used fraction of the cap stays flat at 0.50–0.52, the envelope noise term dominating. This is not a wrong rate law in the theory; it is the honest price of a product-of-sups over correlated quantities, and it is paid where the cap can afford it — the model term stays sub-noise at every rate, so its growth never reaches the observable. The envelope (90), which grows like $\sinh x_{\text{end}}$, bounds e_ω — not the residual — and comparing the residual against it is what made the old cap look loose.

The trend is an anchor choice, and the theory contains the flat anchor too. Why is the realised δe_ω flat while its bound grows? Because the Dirichlet boundary zero sits directly under the $\dot\rho$ peak, the true response is governed not by the peak of $\dot\rho$ (which rides $\omega_{\max}$ and doubles) but by the ρ *level*: $\int \dot\rho = \rho(\tau_{\text{end}}) \propto \varphi^2_{\text{end}}$, flat in rate since the realised end tilt is. A bound anchored on $\rho(\tau_{\text{end}})$ exists — the Green-kernel route of Remark (a), $|\delta e_\omega| \le (1/J_P)\int |K|\rho_{\text{env}}$ — and evaluated per run (`analysis/k_route_trend.py`) it inherits exactly that flatness: 0.29–0.32°/s across the sweep at the 5° box, a trend of ×1.08 against the deployed route's ×1.4. The two routes now cross *inside* the sweep: the ρ-anchored form starts winning on runs at the fastest ramps, and the per-run minimum of the two — valid, since each is a valid bound — runs 0.23–0.28 with trend ×1.24, 140/140, worst used 0.77. The reviewer-facing statement: the full bound is min($\dot\rho$-anchored, ρ-anchored); its growing arm happens to be the binding one on this sweep, and the flat arm — matching the data's trend — takes over beyond it. The growth is which arm binds, not a wrong law.

How far the shift moves the critical moment. The absorbed shift is (9), and the moment error it causes cancels $\dot M$ exactly:

$$\left|\Delta M_{\text{crit}}\right| \;=\; \dot M\,|\delta t_c| \;\approx\; \dot M\cdot\frac{e_\omega(\tau_{\text{end}})}{C_1C_2\sinh x_{\text{end}}} \;=\; \frac{W z_{CoM}\,e_\omega(\tau_{\text{end}})}{C_2\sinh x_{\text{end}}} \;\le\; \frac{W z_{CoM}\,E(\tau_{\text{end}})}{C_2\sinh x_{\text{end}}} \;=\; \bar\rho, \qquad (21)$$

using $\dot M/C_1 = W z_{CoM}$ and $E(\tau_{\text{end}}) = \bar\rho\sinh x_{\text{end}}/(J_P C_2)$. So the shift can displace the identified threshold by at most the window-averaged remainder — the same $\bar\rho$ the manuscript already budgets. At the 5° box the per-run ceiling is 1.8–3.3 mN m; the e_ω-driven part of the shift is guaranteed $\le \bar\rho$ and realised $\le 0.29\,\bar\rho$ (the exact-solution envelope ratio), the sign lowering M_{crit} (earlier onset). The measured onset scatter, $\dot M|\delta t_c| = 0.36$–$1.85$ mN m from the fits, additionally contains noise-driven jitter — comparable to the ceiling in size but governed by the noise term, not by (21). (The chain's middle step is the first-order $|\delta t_c|$; exactly, $|\delta t_c| = \operatorname{artanh}(\beta/C_1)/C_2$, so the ceiling acquires a factor $\operatorname{artanh}(u)/u$: ≤ 1.009 on the campaign envelope, ≤ 1.007 over the design box — negligible at the 5° box.)

10 The bound over the design box

Because every constant in (17), (18) and (21) is available before the experiment, both results can be swept over the vehicle's design box — mass 3.00–3.22 kg, $l_p = 0.110$–0.140 m, CoM offset $p_{\text{off}} = \pm 0.020$ m (arm $= l_p - p_{\text{off}}$), $z_{CoM} = 0.20$–0.30 m, $J_{CoM} = 0.050\ \text{kg m}^2$ with J_P by the parallel-axis theorem, 600 combinations at each of the seven rates, each with its own tilt-limited window ($\sinh x_{\text{end}} - x_{\text{end}} = \varphi_{\max} W z_{CoM} C_2/\dot M$).

Two readings (Fig. 4). The residual bound stays within 0.07–0.25°/s everywhere in the box — an order of magnitude below the measured disturbance floor of ≈ 1°/s at every corner — so the fit-quality gate can be set from the noise figure alone for any vehicle in this class. And the identification error from the absorbed onset shift never exceeds 3.1 mN m anywhere in the box, at any rate — inside its $\bar\rho$ budget everywhere, the exact-shift excess below 0.7% — with the same flatness in $\dot M$ that the campaign shows.

Reproduction

All numbers and figures in this document come from three scripts, run from the repository root:

```
python analysis/failing_runs.py        DataSet/exp   # writes the cache
python analysis/kernel_free_bound.py  docs/fig_kernel_free.png
python analysis/design_range_chart.py docs/fig_design_range.png
python analysis/noise_spectrum_figure.py docs/fig_noise_spectrum.png
python analysis/sg_bound_figure.py    docs/fig_sg_bound.png
```

`kernel_free_bound.py` evaluates (16)–(20) per run against the campaign cache and draws Fig. 3 (`M2()`, `M1()` implement (17a)–(17b); `model_term()` implements (16)–(18)). `design_range_chart.py` sweeps the design

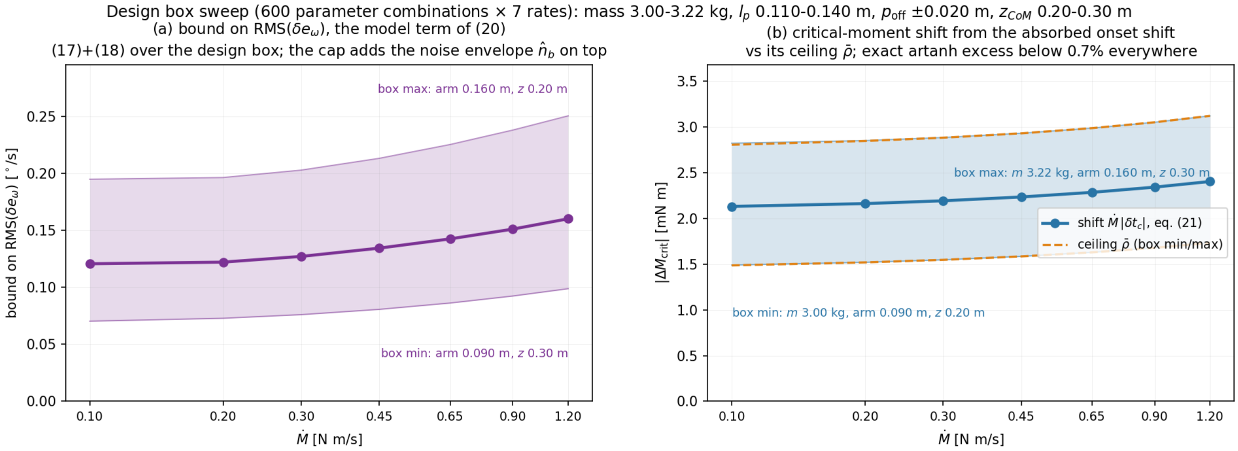

Figure 4: The design-box sweep. *(a)* the bound on RMS(δe_ω), eq. (17)+(18): 0.07–0.25°/s over box and rates, median 0.11–0.16. The band is set by the ratio arm$/z_{CoM}$ — worst at arm 0.160 m with $z_{CoM} = 0.20$ m, best at arm 0.090 m with 0.30 m. *(b)* the critical-moment shift $\dot{M}|\delta t_c|$ of (21): 1.5–3.1 mN m, hugging its ceiling $\bar{\rho}$ — the $\operatorname{artanh}(u)/u$ excess of the exact shift is below 0.7% everywhere at the 5° box — and flat in $\dot{M}$ by the cancellation. Its worst corner has the *opposite* z direction — $m = 3.22$ kg, arm 0.160 m, $z_{CoM} = 0.30$ m — because $\bar{\rho}$ grows with the window that a higher CoM lengthens, while the residual bound worsens with arm$/z_{CoM}$.

box of Sec. 9 with the same closed forms and draws Fig. 4; its box limits are constants at the top of the file. Neither script solves an ODE or builds a kernel matrix — elementary functions and one scalar root-find (the window equation) per combination.

Remarks

(a) Without the differential equation. The same δe_ω can be written directly from the manuscript's Duhamel form (89), giving $\delta e_\omega = \frac{1}{J_P}\int_0^{\tau_{\mathrm{end}}} K(\tau, s)\rho(s)\, ds$ with a kernel that the addition identities collapse to $K = \sinh(C_2(\tau_{\mathrm{end}} - \tau))\cosh(C_2 s)/\sinh(C_2\tau_{\mathrm{end}})$ for $s \le \tau$ (minus its mirror for $s > \tau$), and $\max|K| = 1$ exactly — the exponential cancels inside the kernel. It is an instructive identity, but as a bound it is weaker: K changes sign at $s = \tau$, so only $\int |K|$ can be used, and that costs a factor ≈ 2.4 over (17). The derivative bound in (16) is what forbids ρ from flipping sign with K, and that smoothness is worth exactly that factor.

(b) What is a priori and what is not. The model term of (20) uses only pre-experiment constants (list under (16)). The noise term (19) is measured: n_{hi} from the run itself, κ_q from quiet stretches, s_{med} one campaign median. A fully pre-experiment cap replaces $\hat{n}$ with a bench noise figure.

(c) Exponent mismatch. If the true exponent differs from the pinned C_2 by a relative ε_2, the mismatch joins e_ω by definition and its unabsorbed part costs 0.018–0.022°/s per 1% of ε_2 — 7–10% of the model term, and the calibration residuals limit ε_2 to a few percent.

Appendix: the linear-algebra picture

Collect the N samples of a signal into a vector in $\mathbb{R}^N$. The model family (2) is then a subset of that space,

$$\mathcal{M} \;=\; \Big\{\, a\big(\cosh(c(\tau_i - d)) - 1\big) + b \;:\; a, c > 0,\; d, b \in \mathbb{R} \,\Big\} \;\subset\; \mathbb{R}^N, \tag{A.1}$$

a four-parameter curved manifold. Its tangent space at the nominal is spanned by the four derivative directions

$$T \;=\; \operatorname{span}\{\, \underbrace{\cosh(C_2\tau) - 1}_{\partial/\partial C_1},\; \underbrace{\sinh(C_2\tau)}_{\partial/\partial \delta t_c},\; \underbrace{\tau\sinh(C_2\tau)}_{\partial/\partial C_2},\; \underbrace{\mathbf{1}}_{\partial/\partial C} \,\}, \tag{A.2}$$

so the family can move in at most four of the N dimensions around any of its points.

Why exact representation fails. "The true curve is a shifted nominal" asserts membership: $y_{\mathrm{struct}} \in \mathcal{M}$. Two obstructions, matching the two in the Sec. 4 remark. Dimension: a four-dimensional manifold is a measurezero sliver of $\mathbb{R}^N$; a vector off it stays off it. Dynamics, sharper: every element of $\mathcal{M}$ samples a homogeneous solution of $\ddot{v} = c^2 v$, while y_{struct} samples the forced equation (12); by the argument under (13), membership would force $\dot{\rho} \equiv 0$. So y_{struct} has a component *transverse* to $\mathcal{M}$ that no parameter choice reaches.

What introducing δe_ω does. It changes the assertion from membership to a coordinate statement:

$$y_{\text{struct}} \;=\; m \;+\; \delta e_\omega, \qquad m \in \mathcal{M}, \tag{A.3}$$

exact for *every* choice of m — the freedom is in where to put the remainder, not whether there is one. Two canonical choices divide the labour:

choice of m	property of δe_ω	what it buys
orthogonal projection (LS)	$\delta e_\omega \perp T$, minimal norm	the smallest remainder, no pointwise structure
end-matched witness, (9)	$\delta e_\omega = 0$ at both window ends	the Dirichlet problem (13)–(14), hence (17)

The witness pays a factor 1.8–2.0 in norm over the orthogonal choice and buys, in exchange, the two boundary zeros that make the maximum principle — and with it the entire a priori bound — available. Provability is purchased with norm.

The projector sub-case, and the manuscript's Appendix C. With C_2 pinned and the onset held, the family linearises to the subspace $\mathcal{V} = \text{span}\{u, \mathbf{1}\}$, $u_i = \cosh(C_2\tau_i) - 1$, and the orthogonal choice becomes an explicit matrix: with $A = [\,u \;\; \mathbf{1}\,]$,

$$P = A(A^\top A)^{-1}A^\top, \qquad P_{ij} = \frac{1}{N} + \frac{\tilde{u}_i\tilde{u}_j}{\sum_k \tilde{u}_k^2}, \qquad r_\perp = (I-P)(e_\omega + n), \tag{A.4}$$

($\tilde{u} = u - \bar{u}$) — the manuscript's Lemma 2, eq. (C3). The δe_ω of this document is the same object one level up: the remainder transverse to the full nonlinear family rather than to its pinned linearisation.

How much of e_ω lies along the family. Measured on the exact tip-over, the fraction of e_ω (in RMS) absorbed by successively larger reachable sets: 61–67% by span$\{u, \mathbf{1}\}$, 70–71% adding the onset direction sinh, 92.5% by the full four-parameter family — the last flat across all rates. The transverse component — the 7.5% no parameters reach, 0.012–0.018°/s — is the irreducible core of δe_ω, and (17) is its a priori ceiling.